

Multi-Asset Portfolio Risk Decomposition and Stress Testing: An Open-Source Institutional-Style Framework Using ETF Proxies

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Abstract

I build an open-source risk analytics framework that treats a multi-asset portfolio the way an institutional risk desk does, asking what risks it contains and how it behaves when markets break rather than whether it earns alpha. Using eleven liquid ETF sleeves spanning global equities, Treasuries, credit, gold, real estate and cash over 2007 to 2026, it delivers exact Euler risk decompositions, an observable six-factor model, stress tests with a model-validation step, regime-conditional covariance, and cost-aware walk-forward construction. Applied to a 60/40 portfolio, equity sleeves supply 51% of capital but 84% of risk; the Treasury hedge that justified the defensive sleeve has weakened as the stock–bond correlation turned positive after 2022; and the portfolio’s binding stress scenario is consequently an inflation shock rather than an equity crash. A constrained, low-turnover construction recovers most of the lost diversification at a Sharpe cost within estimation noise, and it keeps that edge net of realistic trading costs while an unconstrained maximum-Sharpe portfolio’s edge erodes with size. Everything is reproducible from free data.

1 Introduction

Most student research portfolios answer an alpha question. This note answers a fiduciary one. The distinction matters because the two lenses look at the same return series and see different objects: the alpha lens sees a Sharpe ratio, while the risk lens sees a bundle of factor exposures, a covariance that shifts with the regime, and a set of scenarios under which a client loses money that they were told was diversified away. The headline result states the change of lens concretely. The classic 60/40 is capital-diversified but far less risk-diversified than its allocation suggests, with equity sleeves supplying 51% of capital and 84% of volatility, and its binding stress scenario has shifted from an equity crash to an inflation shock since the stock–bond correlation turned positive in 2022.

The framework behind that diagnosis, `multi-asset-risk-lab`, is a deliberately institutional exercise built entirely from free, reproducible data. It mirrors the workflow of a commercial risk platform: map holdings to sleeves, decompose risk so that every table reconciles exactly, express exposures in factor language, validate the factor model before trusting it, condition risk on the macro regime, stress the book against history and hypotheticals, and only then ask the construction question, net of what it costs to trade. Nothing here proposes a new optimiser or risk model; every technique is standard on a risk desk. The contribution is the integrated open-source workflow itself, which shows how risk decomposition, factor exposure, regime conditioning, stress testing, implementation cost and attribution successively alter the diagnosis of the same portfolio, with exact reconciliation and the conventions stated inline.

Three results carry the narrative. Capital weights turn out to be a poor description of risk: the 60/40's 40% defensive allocation carries almost none of its volatility. Risk models estimated on full-sample covariances misstate the current regime, because the stock–bond correlation, negative for the entire 2008 to 2021 period, has been persistently positive since 2022, converting the Treasury allocation from a free hedge into a risk asset. And the scenario ranking of the portfolio inverts accordingly: once the defensive sleeve stops hedging, an inflation shock becomes the binding case. The later sections show that these conclusions survive when risk is conditioned on regime, when construction is compared out of sample, and when turnover is charged at a realistic cost.

2 Data and universe

The universe is eleven US-listed ETFs, one per broad exposure: SPY, VGK, EWU and EEM for US, European, UK and emerging-market equity; TLT and IEF for long and intermediate Treasuries; LQD and HYG for investment-grade and high-yield credit; GLD for gold; VNQ for listed real estate; and BIL as the cash proxy. Prices are total-return adjusted closes from Yahoo Finance, cached locally with daily share volumes so the full pipeline runs offline.

Because HYG and BIL only launched in 2007, the common full-panel sample runs from 4 June 2007 to July 2026 (about 4,800 trading days), reported as such rather than stretched to the earliest sleeve's history. The sample still spans six named stress windows

Table 1: Portfolios compared in the note and the idea each represents.

Portfolio	What it represents
Equal weight	Naive $1/N$ diversification across sleeves; a capital-diversification benchmark.
60/40	The textbook balanced mandate; the note’s main subject.
Minimum variance	Lowest-volatility long-only portfolio on the sample covariance.
Min variance (LW)	The same, on a Ledoit–Wolf shrinkage covariance; the better-conditioned default.
Risk parity	Equal risk contributions across sleeves; diversifies risk, not capital.
Maximum Sharpe	Unconstrained mean–variance optimum; the estimation-error stress case.
Minimum CVaR	Minimises empirical tail loss; a distribution-free alternative.
Client-friendly	Maximum Sharpe inside IPS-style caps and asset-class ranges.

used throughout: the global financial crisis, the 2011 euro sovereign crisis, the 2013 taper tantrum, the Q4 2018 selloff, the COVID crash, and the 2022 rate shock.

Table 1 lists the portfolios the note compares and what each is meant to represent. Two baselines anchor the analysis. *Equal weight* places 1/11 in each sleeve. The *60/40* maps the textbook balanced mandate onto the universe: 60% growth (36% SPY, 6% VGK, 3% EWU, 6% EEM, 5% VNQ, 4% GLD) and 40% defensive (16% IEF, 12% TLT, 12% LQD), rebalanced monthly. Over the common sample the 60/40 returned 7.6% annualised at 11.3% volatility (Sharpe 0.67) with a -33.5% maximum drawdown. The remaining portfolios are constructed in Section 7.

3 Risk decomposition

3.1 Methodology

All decompositions are Euler decompositions, so components sum to the portfolio total exactly and every table reconciles. With weights w and covariance Σ , portfolio volatility is $\sigma_p = \sqrt{w'\Sigma w}$, the marginal risk contribution of sleeve i is $\text{MRC}_i = (\Sigma w)_i / \sigma_p$, and its component volatility $w_i \text{MRC}_i$ sums to σ_p . Parametric value-at-risk uses the zero-mean normal convention standard for short horizons, so component VaR inherits the Euler property. Historical expected shortfall is decomposed by averaging each sleeve’s realised contribution on tail days; because daily contributions are built from actual drifted weights and sum to the daily portfolio return, component ES sums to portfolio ES to machine precision. Risk is measured under two covariances: the full-sample estimate, and a RiskMetrics-style exponentially weighted estimate ($\lambda = 0.94$) that tracks the current regime.

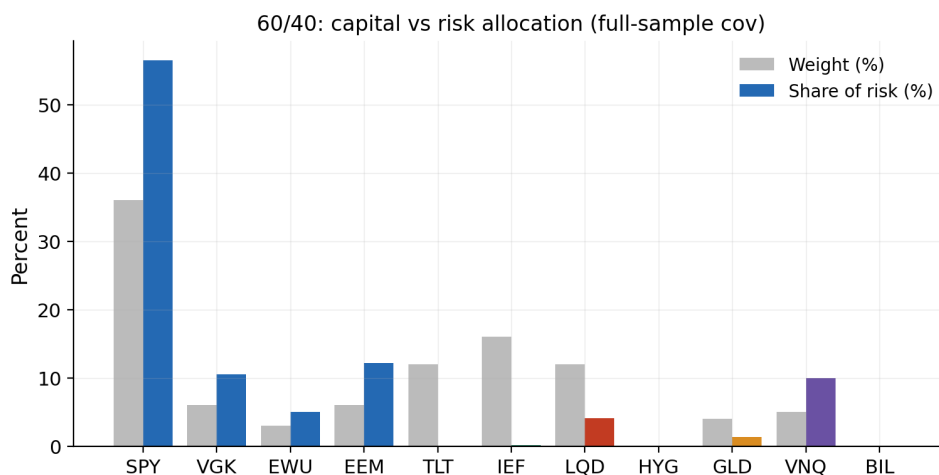


Figure 1: Capital versus risk allocation for the 60/40 under the full-sample covariance. Grey bars are portfolio weights; coloured bars are Euler shares of portfolio volatility.

3.2 Capital weights understate equity risk

Figure 1 shows the headline result. Equity sleeves hold 51% of the 60/40's capital but contribute 84% of its volatility, and SPY alone turns 36% of capital into 57% of risk. The 60/40 remains capital-diversified but is markedly less risk-diversified than its headline allocation implies. The pattern is not specific to the 60/40: even the equal-weight portfolio, with only 36% in equity sleeves, draws 67% of its risk from them.

3.3 The hedge that weakened

The more consequential result concerns what the other 49% of capital is doing. Under the full-sample covariance, TLT contributes 0.03% of the 60/40's risk from a 12% weight, an essentially free hedge, which is exactly what justified the allocation. Under the exponentially weighted covariance the same sleeve contributes 6.4%, and the two Treasury sleeves together 12%. Figure 2 shows why: the 126-day stock–bond correlation, negative for virtually the entire 2008 to 2021 period, flipped positive during the 2022 inflation shock and has remained predominantly positive since. A risk model calibrated to the full sample tells the 2007 to 2021 story, but the portfolio lives in the post-2022 one.

3.4 Drawdown attribution

The worst 60/40 drawdown (−33.5%, October 2007 to March 2009) attributes cleanly: SPY contributed −25.9pp, European equity −5.4pp and emerging markets −4.8pp, while the Treasury sleeves added roughly +3pp each (Figure 3). The attribution sums to the arithmetic window return, with the +4.1pp compounding cross-term reported separately rather than smeared across sleeves. The GFC is the regime in which the 40 worked. The 2022 window is the regime in which it did not, and Section 5 shows that there every sleeve except gold and cash lost money at once.

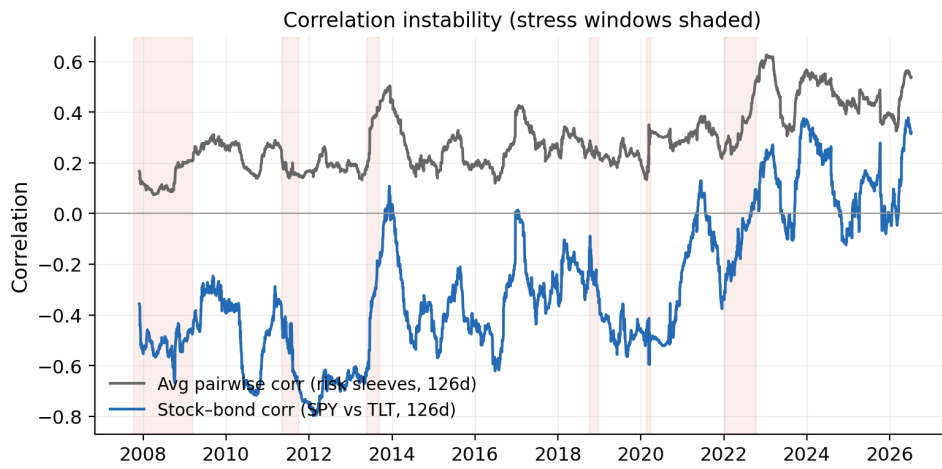


Figure 2: Rolling 126-day average pairwise correlation across risk sleeves (grey) and the SPY–TLT stock–bond correlation (blue). Shaded bands mark the six named stress windows.

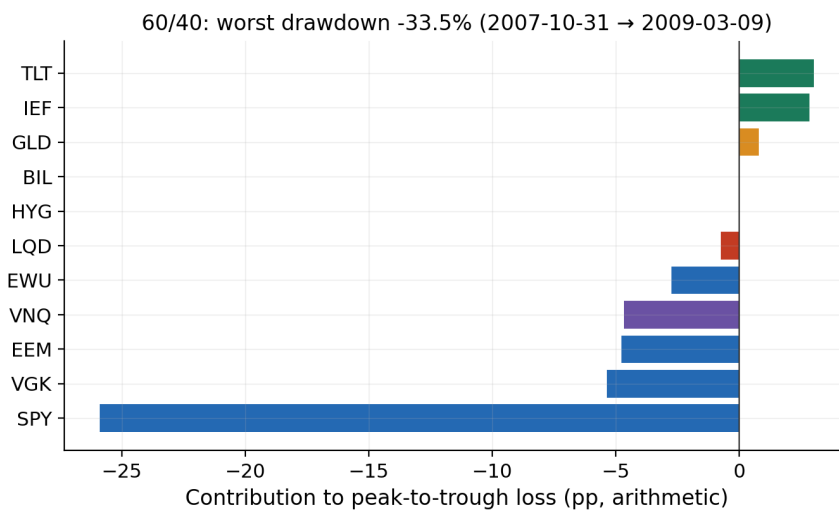


Figure 3: Attribution of the 60/40's worst drawdown. Bars are per-sleeve contributions to the arithmetic peak-to-trough loss.

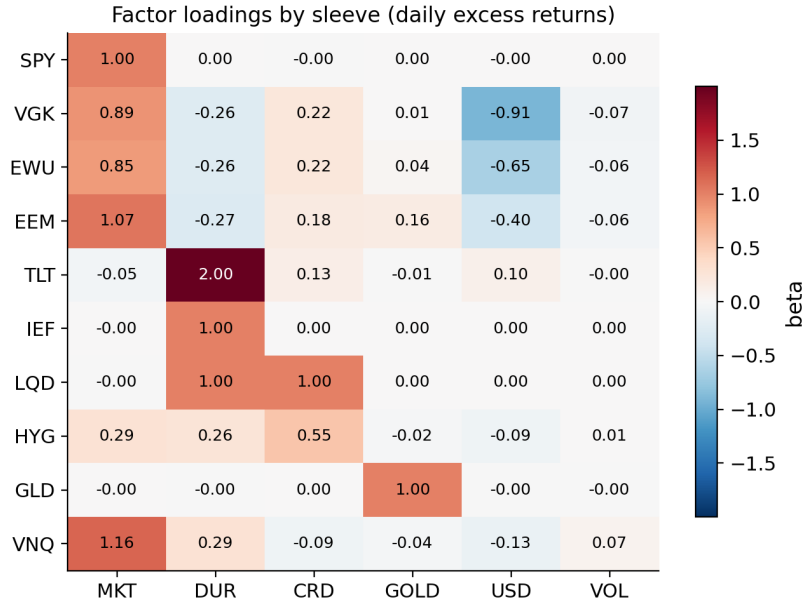


Figure 4: Full-sample factor loadings by sleeve, daily excess returns, 2007–2026. Rows for SPY, IEF, LQD and GLD are identification checks (these sleeves are factor constituents); the remaining rows are genuine estimates.

4 An observable factor model

4.1 Construction

Ticker-level risk tables answer “which line item,” not “which risk.” The factor model re-expresses the book in six observable, tradeable factors: market (SPY in excess of cash), duration (IEF in excess of cash), credit (LQD minus IEF, duration-hedged and self-financing), gold, dollar (UUP in excess of cash), and volatility. The volatility factor is the daily VIX change orthogonalised against the market factor, since raw VIX changes correlate about -0.7 with equity daily and an unorthogonalised vol beta would double-count market exposure. Assets are regressed in excess of cash with Newey–West standard errors (five lags), because stale pricing in credit ETFs induces enough autocorrelation to overstate plain OLS precision.

4.2 Validation

The model validates itself in ways not imposed by construction (Figure 4). LQD loads 1.00 on duration and 1.00 on credit with $R^2 = 1$, confirming the spread factor reconciles exactly. TLT’s duration beta is 2.00, the duration ratio of 20-year to 7–10-year Treasuries. Unhedged European equity carries a -0.91 dollar beta, UK equity -0.65 , emerging markets -0.40 ; high yield decomposes into 0.29 equity, 0.55 credit and 0.26 duration, the textbook hybrid.

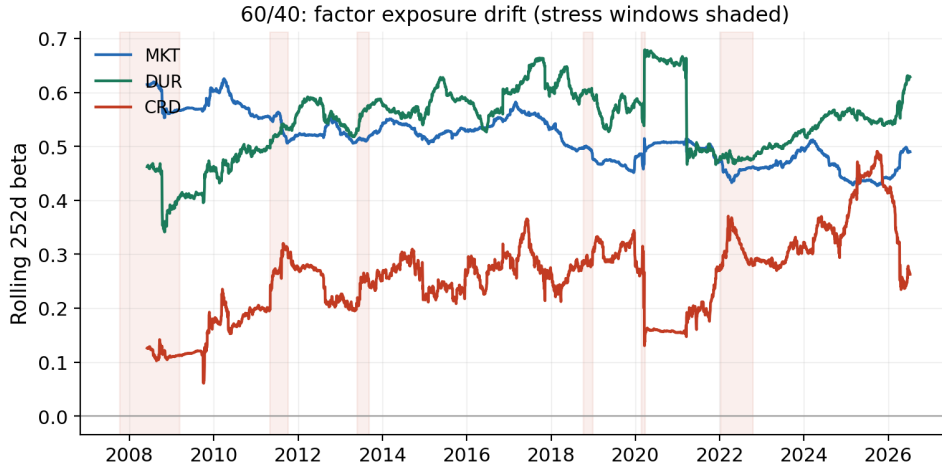


Figure 5: Rolling 252-day market, duration and credit betas of the monthly rebalanced 60/40. Shaded bands mark stress windows.

4.3 Portfolios in factor language

Six factors explain 97% of the 60/40's daily variance ($R^2 = 0.97$, idiosyncratic volatility 2.1%), and the market factor alone accounts for 86% of its risk. The equal-weight portfolio carries a -1.8% annualised alpha against the factor set ($t = -2.5$): its non-US sleeves added factor exposure but subtracted selection, a reminder that diversifying tickers is a weaker exercise than diversifying risk premia.

Rolling 252-day betas (Figure 5) show that fixed weights do not deliver fixed exposures. The 60/40's duration beta ranged from 0.34 to 0.68 across the sample and sits near its highs today, precisely when duration no longer hedges.

5 Stress testing

5.1 Historical replay

Figure 6 reports each sleeve's compounded return through the six stress windows. The TLT row is the note's thesis in one line: $+25\%$ through the GFC, $+35\%$ through the euro crisis and $+14\%$ through COVID, then -31% through 2022. The hedge worked in every crisis until the crisis was inflation.

Replaying today's 60/40 weights through the GFC loses -26.0% , while the monthly-rebalanced portfolio actually experienced -32.6% peak-to-trough, because rebalancing buys a falling market on the way down and earns the difference back in the recovery. Reporting both numbers separates the instantaneous risk of the current book from the path-dependence of the strategy.

5.2 Validating the model before trusting it

Hypothetical scenarios lean entirely on the factor model, so the model must first earn that right on realised history. For each window I compare the beta-predicted arithmetic

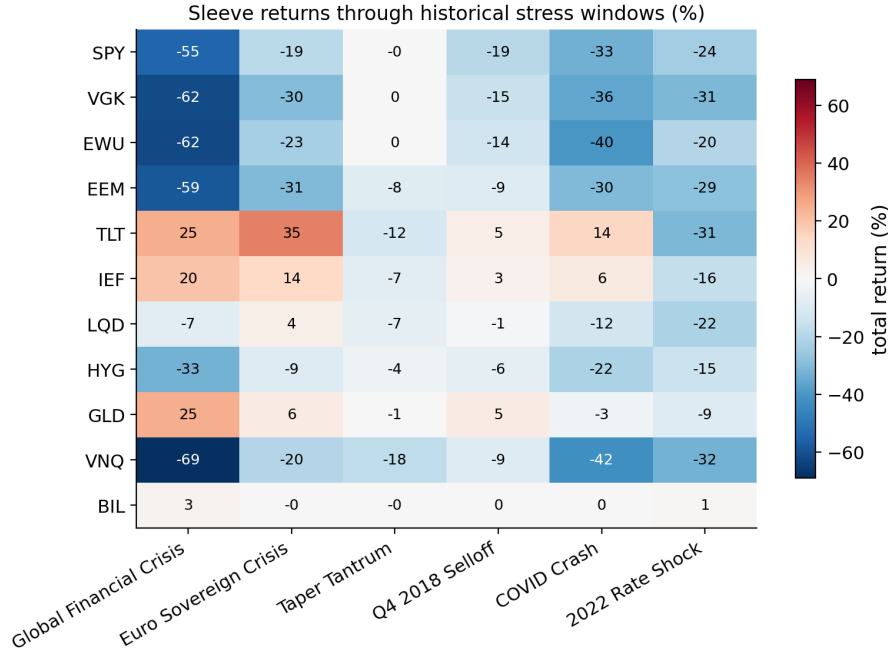


Figure 6: Sleeve total returns through the six named stress windows.

portfolio return against the realised one. Mean absolute error is 1.4% for the 60/40 and 2.1% for equal weight, on windows with 20 to 40% moves, with known weak spots reported rather than hidden: the model under-predicts the taper tantrum and over-predicts Q4 2018 by roughly 2 to 4pp, both periods where idiosyncratic and non-US dynamics dominated.

5.3 Hypothetical scenarios

Seven scenarios are defined in factor language and propagated through the loadings at first order (Figure 7). Rate shocks translate via stated duration assumptions (+100bp gives DUR -7.5% via IEF's 7.5-year duration; +200bp IG widening gives CRD -16% via LQD's 8-year spread duration).

The ranking is the result. The 60/40's worst hypothetical is the stagflation shock (-13.0%), narrowly ahead of the -25% equity risk-off (-12.5% , cushioned by its 0.50 duration beta), because in a stagflation the same duration beta fires in the same direction as equity. The equal-weight portfolio, which carries less duration, ranks the two the other way round. Full-sample volatility cannot see this distinction, but scenario language can.

6 Regime-conditional risk

Full-sample and even exponentially weighted covariances describe an average world. A portfolio's risk identity is not constant: the same 60/40 that is duration-hedged in a flight to quality is duration-exposed in an inflation shock. To make this explicit I tag each day with a coarse macro regime from market signals alone (trailing equity trend and realised volatility, the stock–bond return sign, and high-yield performance relative to investment grade), then re-estimate risk conditional on the regime. The five regimes are

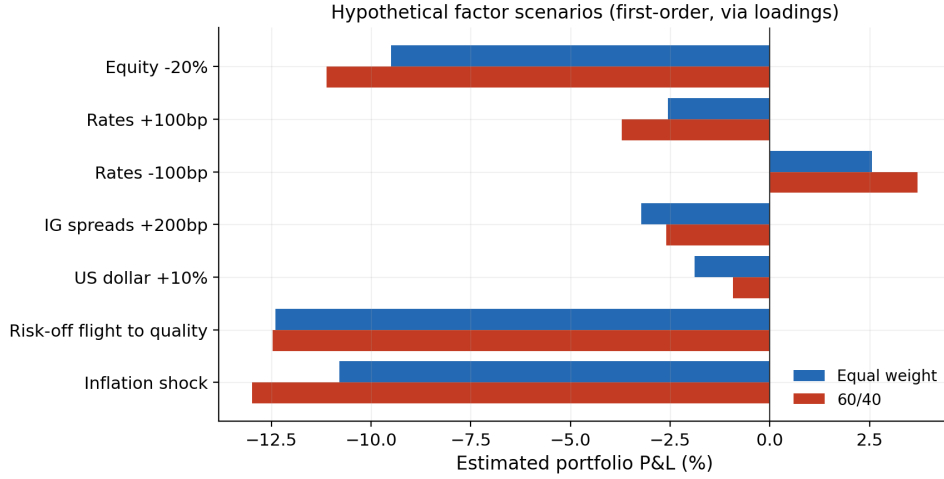


Figure 7: First-order scenario P&L for the baselines, via full-sample loadings.

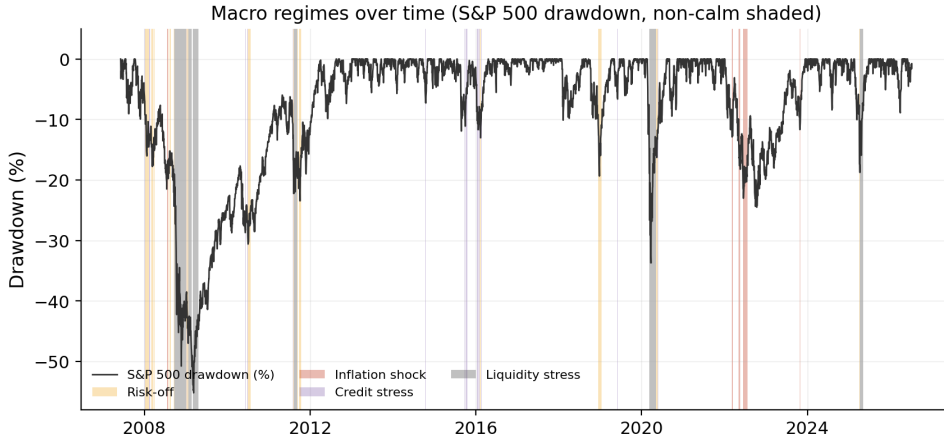


Figure 8: Macro regimes over time, drawn against the S&P 500 drawdown. Non-calm regimes are shaded.

calm growth, risk-off, inflation shock, credit stress and liquidity stress, assigned by fixed rules in priority order so the more acute state wins. Figure 8 shows that the labels line up with the episodes one would draw by hand: the 2008 and 2020 troughs read as liquidity stress, the 2011 and 2018 selloffs as risk-off, and the 2022 drawdown as an inflation shock.

Table 2 conditions the 60/40 on each regime. The same weights behave like a different portfolio in each column: annualised volatility ranges from 8.7% in calm growth to 34.4% in liquidity stress, a fourfold span, and the equity share of risk moves from 79% to 97%. In credit stress the equity risk share exceeds 100% because the bond sleeves carry a small negative risk contribution there, hedging rather than adding. A single covariance cannot represent a book that changes shape this much, which is the case for holding several estimates side by side rather than one.

Table 2: The 60/40’s risk identity conditional on macro regime. Volatility and equity risk share use the regime-conditional covariance; returns are noisy in the rare regimes and are shown for context.

Regime	Days	Ann. vol	Mean ann. return	Equity risk share
Calm growth	88.4%	8.7%	+10.8%	79%
Risk-off	3.8%	16.1%	−53.0%	97%
Inflation shock	1.4%	14.0%	−16.6%	83%
Credit stress	1.9%	11.1%	+12.2%	103%
Liquidity stress	4.5%	34.4%	+13.8%	87%



Figure 9: Full-sample weights by construction scheme, coloured by asset class.

7 Portfolio construction

Six schemes are compared against the baselines: long-only minimum variance on the sample covariance and on a Ledoit–Wolf shrinkage covariance, equal-risk-contribution risk parity (log-barrier formulation, with realised risk contributions equal to within 2×10^{-5}), unconstrained maximum Sharpe consuming plain historical mean excess returns, minimum CVaR via the Rockafellar–Uryasev linear program, and a client-friendly maximum Sharpe inside IPS-style limits (25% sleeve cap; equity 35 to 60%; rates at least 15%; credit at most 25%; gold and real estate at most 10% each). Optimisation runs over the ten risky sleeves, since including cash would let minimum variance and risk parity degenerate into T-bill funds, and cash is a sizing decision rather than an allocation. Figure 9 shows the full-sample weights.

The shrinkage covariance is worth a comment. With about 4,800 observations and ten assets, the sample covariance is already well estimated, so the Ledoit–Wolf intensity is only 0.008 in-sample and the minimum-variance weights it produces are within a percentage point of the plain estimate. Shrinkage earns its keep when assets are many relative to observations, which the three-year refit windows below approach more closely than the full sample does. Reporting that it barely moves the answer here is more useful than

Table 3: Walk-forward out-of-sample performance, January 2011 to July 2026. Annual refit, three-year lookback, monthly rebalancing to target. Turnover is one-way per annual refit; cost is the annualised drag from ongoing refit trades at \$1bn assets.

	Net ret	Vol	Gross SR	Net SR	Turnover	Cost bp
Equal weight	6.3%	8.9%	0.71	0.71	–	–
60/40	8.3%	9.6%	0.87	0.87	–	–
Min variance	4.0%	5.3%	0.76	0.75	12%	3
Min variance (LW)	4.0%	5.3%	0.76	0.75	11%	3
Risk parity	5.8%	7.3%	0.79	0.79	6%	1
Max Sharpe	9.3%	9.6%	0.99	0.97	49%	20
Min CVaR	4.1%	5.4%	0.76	0.76	12%	3
Client-friendly	7.4%	8.2%	0.91	0.90	23%	10

pretending it transformed it.

7.1 Out-of-sample performance, net of cost

Evaluation is strictly walk-forward: each scheme refits annually on a trailing three-year window and holds the weights out of sample, so schemes that consume expected returns face the same estimation error a live desk would. Costs are charged on the change in target weights at each refit through a square-root market-impact model (half-spread plus $k\sigma\sqrt{\text{notional}/\text{ADV}}$) calibrated to each sleeve’s dollar volume, at a default \$1bn of assets. The headline convention charges ongoing refit turnover only, so the initial buy-in is free for optimised schemes and fixed-weight baselines alike and the comparison is symmetric; also charging the day-one transition (the mandate-transition convention) lowers net Sharpe by at most 0.01 and changes no ranking. Table 3 reports gross and net results for 2011 to 2026.

Several judgements follow. Maximum Sharpe leads out of sample, and the number should be treated with caution: it churned 49% of the book at each refit and held zero non-US equity or credit for fifteen years, winning by concentrating in the era’s winners (SPY, IEF, GLD). That is a bet on the stability of estimation windows, and the scenario engine prices what it would lose if the regime turned. The constrained client-friendly portfolio achieves a 0.90 net Sharpe with half the turnover, enforced diversification and no sleeve above 25%, which for a fiduciary is the better trade. Minimum variance, whether on the sample or the shrinkage covariance, is duration in disguise: 70% IEF produces the lowest historical volatility of any scheme and, from Figure 7’s logic, the worst +100bp rate scenario. Minimum CVaR lands close to minimum variance in this universe because the same low-volatility sleeves also have the thinnest tails, but it reaches them without assuming normality. Low volatility measured on yesterday’s covariance is not the same as low risk in an inflation regime.

Table 4: Net Sharpe as a function of assets under management. Fixed-weight baselines carry no refit turnover and are size-invariant.

	\$1bn	\$5bn	\$10bn	\$25bn
Risk parity	0.79	0.79	0.79	0.78
Client-friendly	0.90	0.89	0.88	0.85
Max Sharpe	0.97	0.94	0.92	0.88

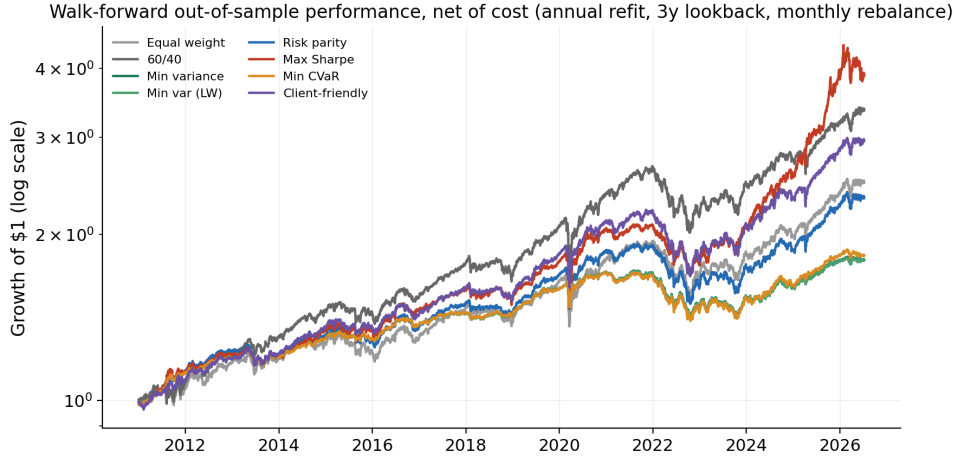


Figure 10: Walk-forward out-of-sample growth of \$1 net of cost, log scale.

7.2 Costs at scale

At \$1bn the cost differences are real but small, because ETF liquidity is deep. Their importance grows with size. Table 4 recomputes net Sharpe as assets rise. Maximum Sharpe’s advantage over the client-friendly portfolio falls from 0.07 at \$1bn to 0.03 at \$25bn, as its turnover is taxed by market impact, while the low-turnover schemes barely move. The lesson for an allocator is that a headline Sharpe earned through high turnover is partly an artefact of assuming costless trading, and the constrained portfolio’s case strengthens precisely at the scale an institution runs.

8 Performance attribution

Attribution uses monthly buy-and-hold returns with weights reset at each month start (the implementable monthly-rebalanced portfolio) and links periods with the Carino method, so every table reconciles to the compounded return with no residual row, at machine precision.

Figure 11 shows the 60/40’s annual contributions by asset class. The two loss years tell opposite stories. In 2008 equity cost -22.7pp while rates added $+5.9\text{pp}$; in 2022 every class was negative at once, with equity -8.5pp , rates -6.4pp and credit -2.2pp . The diversification the client was sold existed in the first crisis and failed in the second.

A Brinson–Fachler decomposition against the 60/40 benchmark completes the picture. Equal weight underperformed by -83.0pp compounded over the sample, of which -83.2pp

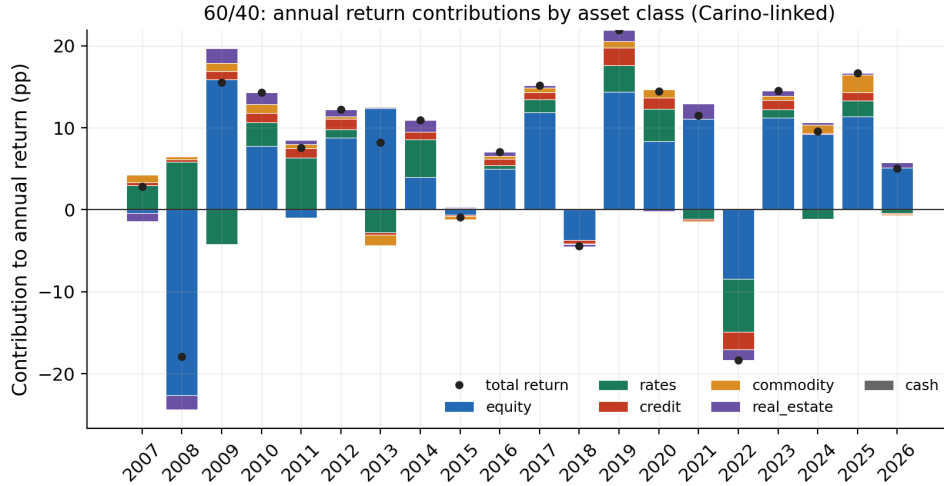


Figure 11: Carino-linked annual return contributions by asset class for the 60/40. Dots mark total annual returns.

is equity selection, since equal-weighting within equity is a large implicit bet on non-US markets that spent fifteen years losing to SPY, and -28.1pp is the standing cash allocation the benchmark does not hold. The client-friendly portfolio's -35.3pp active return is almost entirely allocation (-33.2pp from rates, -22.4pp from equity underweight, partly offset by $+13.9\text{pp}$ from gold), consistent with a risk-driven rather than selection-driven mandate.

9 Investment recommendation

On the evidence here, an allocator holding the 60/40 should reduce its reliance on long duration as an equity hedge and move part of the defensive sleeve towards a constrained, risk-balanced allocation. The client-friendly portfolio delivers essentially the same net-of-cost Sharpe as the more concentrated alternatives while capping single-sleeve and asset-class exposures, and it does not depend on a stock-bond correlation that the data since 2022 no longer support. The risk to flag to a portfolio manager is specific: the current 60/40 is more exposed to an inflation or rate shock than to an equity-only selloff, its effective duration sits near a fifteen-year high, and its volatility roughly quadruples in a liquidity-stress regime. None of these are visible in full-sample volatility or a headline Sharpe ratio, which is the reason to run the decomposition, the regime conditioning and the scenarios before signing off on the allocation.

10 Limitations

The framework's honesty about conventions is part of the point, so the limitations are stated plainly. Scenario propagation is linear and first-order, with no convexity, no idiosyncratic moves and no gap risk. Factor loadings used for hypotheticals are full-sample estimates, and the rolling betas show they drift. Parametric VaR uses the zero-mean

normal convention and understates tails relative to the historical ES reported beside it. The transaction-cost model is a calibrated square-root approximation with no quote data, and it charges refit trades but not intra-year drift, so the reported costs are a lower bound; it is a fair cross-scheme comparison rather than a precise cost forecast. The regime classifier is a descriptive, in-sample labelling with fixed thresholds, not a tradeable timing signal, and its trailing windows mean a label is only known with a lag. ETF proxies embed fees, tracking error and, for credit, stale-pricing smoothing, and the pre-2007 history of several asset classes is unavailable in this universe. Finally, the walk-forward comparison covers a single fifteen-year path dominated by US asset outperformance, so the maximum-Sharpe result in particular should be read as one draw rather than an expectation.

11 Conclusion

Assembled end to end, the framework supports a conclusion that no single statistic delivers. The classic 60/40 currently carries near-record effective duration into a regime where duration no longer hedges equity, which makes an inflation shock, rather than an equity crash, its binding risk scenario, and that conclusion holds when risk is conditioned on regime and survives realistic trading costs. A constrained, risk-parity-tilted construction buys back most of the diversification at a Sharpe cost within estimation noise and keeps its edge net of cost as assets grow. The broader point is that capital weights, full-sample covariances and headline Sharpe ratios are each, on their own, misleading descriptions of a multi-asset portfolio. Risk decomposition, regime-aware covariance, validated scenarios, cost-aware construction and reconciling attribution are what turn a return series into a decision.

A Reproducibility

The framework is a Python package (`risklab`) with 64 unit tests covering the Euler identities, factor recovery, the transaction-cost scaling law, CVaR optimality, the Ledoit–Wolf conditioning improvement, the regime rules, and an offline end-to-end run of the whole pipeline. Prices and volumes are total-return adjusted closes and share volumes from Yahoo Finance, cached one file per ticker; a committed synthetic fixture set lets the entire workflow run with no network access. Every figure and table in this note is produced by a script under `scripts/` (`run_risk`, `run_factors`, `run_stress`, `run_regimes`, `run_construction`, `run_attribution`). Dependencies are pinned in `requirements.txt`, and a `Makefile` exposes `make data`, `make reports` and `make test`.

Code and reproduction: all data pipelines, risk engines, tests and figure scripts are in the accompanying repository (`multi-asset-risk-lab`). Prices are total-return adjusted closes from Yahoo Finance. Nothing in this note is investment advice.

References

- Amihud, Y. (2002). Illiquidity and stock returns: Cross-section and time-series effects. *Journal of Financial Markets*, 5(1), 31–56.
- Brinson, G. P., and N. Fachler (1985). Measuring non-US equity portfolio performance. *Journal of Portfolio Management*, 11(3), 73–76.
- Carino, D. R. (1999). Combining attribution effects over time. *Journal of Performance Measurement*, 3(4), 5–14.
- DeMiguel, V., L. Garlappi, and R. Uppal (2009). Optimal versus naive diversification: How inefficient is the $1/N$ portfolio strategy? *Review of Financial Studies*, 22(5), 1915–1953.
- Ledoit, O., and M. Wolf (2004). Honey, I shrunk the sample covariance matrix. *Journal of Portfolio Management*, 30(4), 110–119.
- Maillard, S., T. Roncalli, and J. Teiletche (2010). The properties of equally weighted risk contribution portfolios. *Journal of Portfolio Management*, 36(4), 60–70.
- Rockafellar, R. T., and S. Uryasev (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41.
- RiskMetrics Group (1996). *RiskMetrics: Technical Document*, 4th edition. J.P. Morgan/Reuters, New York.